The late losses of the mixture fit: theory, predictions and the experiments that decide

Version-4 basket model, branch `v4-mixture`

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The mixture form (`docs/mixture_theory.pdf`) did what it was introduced for: one posterior peak per trip, exact generation, a reliable likelihood estimate, and a better held-out score than the old form. This note is about the *training*. After its best step the fit loses held-out likelihood, on synthetic and on real data, and its generated baskets drift in size. Section 1 fixes notation. Section 2 lists what has been measured. Sections 3–6 take each mechanism that could cause the losses, say what it predicts, and say which measurement already decides it. Section 7 is the experiment plan, with the result each mechanism predicts stated before any run.

1 Setting and notation

Data. Trips t, each with a basket S_t and a household $h(t)$. The fit uses T training trips: $T = 7{,}695$ in the synthetic world `disc_c48_s3` (928 households, about 8 trips each) and $T = 158{,}983$ on Dunnhumby (1,920 households, about 83 trips each). 1,024 further trips are held out.

The model. A trip draws a taste $z \in \mathbb{R}^r$ ($r = 16$) from $\mathcal{N}(0, I)$, then a basket from the exact law given z:

$$p(S) = \int \mathcal{N}(z; 0, I)\, p(S \mid z)\, dz, \qquad p(S \mid z) = \frac{e^{E_z(S)}}{Z_z}. \tag{1}$$

For a basket S with n_c products in category c and k categories present,

$$E_z(S) = \sum_{j \in S} \Big(b_j + \lambda_j - \tfrac{1}{2}|\varphi_j|^2 + z \cdot \varphi_j \Big) - \sum_c (n_c - 1)_+ \Big(z \cdot \psi_c - \tfrac{1}{2}|\psi_c|^2 \Big) - \sum_c \big[\rho_c \tbinom{n_c}{2} + f_c(n_c) \big] - \rho_0(k). \tag{2}$$

The symbols: b_j is product j's utility from the household, context and price terms; λ_j is its intercept; $\varphi_j \in \mathbb{R}^r$ is its taste loading; $\psi_c \in \mathbb{R}^r$ is the loading charged for each *extra* product of category c; ρ_c, f_c penalise several products of one category; $\rho_0(k)$ is the size term, indexed by the number of categories k. Z_z normalises over all baskets.

Parameter blocks. *Household*: θ_h, γ_h (40 numbers per household). *Product context*: the per-product context and price factors ($\alpha, \beta, \mu, \zeta, \dots$). *Interaction*: φ, ψ, λ. *Size*: ρ_0, ρ_c, f.

The training procedure (`scripts/fit_mixture.py`). Each step takes a batch of 256 training trips and does the following.

1. It finds each trip's posterior peak, draws $K = 8$ tastes $z_1, \dots, z_K$ from a proposal q, and forms the importance bound

$$L_K(S) = \log \frac{1}{K} \sum_{k=1}^{K} w_k, \qquad w_k = \frac{\mathcal{N}(z_k; 0, I)\, p(S \mid z_k)}{q(z_k)}. \tag{3}$$

   The proposal is $q = 0.8\,\mathcal{N}(m, 1.2^2\Sigma) + 0.2\,\mathcal{N}(m, I)$, where m is the peak and Σ the Laplace covariance. It is deliberately wider than the posterior.

2. It takes one Adam step on $-\overline{L_K}$ + prior$/T$ for every block.

Every 30 steps it also does the following.

3. *Moment matching*: an extra step $\lambda_j \mathrel{-}= 0.35\,\bar{g}_j/r_j$, and the same for ρ_0, ρ_c with factor 0.5. Here $\bar{g}$ is the gradient averaged over the 30 steps and r the data rate.

4. *Whitening*: it estimates the pooled posterior second moment $M = \frac{1}{B}\sum_t \mathbb{E}[zz^\top \mid S_t]$ on 256 fixed trips ($K = 256$). With $A = M^{1/2}$ it sets $\varphi_j \leftarrow A\varphi_j$, $\psi_c \leftarrow A\psi_c$ and $\lambda_j \mathrel{+}= \frac{1}{2}(|A\varphi_j|^2 - |\varphi_j|^2)$. Then it clears Adam's state for φ, ψ.

Held-out score means the mean of $\log p(S)$ over 1,024 held-out trips with $K = 256$. *Training score* means the same over 1,024 training trips. All comparisons are paired, trip by trip, with the same random numbers.

2 What has been measured

	synthetic (step 420 → 1,210)	real (step 360/240 → 1,244)
held-out score	-0.18 ± 0.04	-0.66 ± 0.08 (from step 240)
training score	$+1.31 \pm 0.04$	-0.56 ± 0.06 (from step 240)
generated size (data)	8.68 → 7.92 (8.69)	8.0–8.3 → 7.5–7.9 (7.26)
$\frac{1}{2}\lvert\varphi_j\rvert^2$, mean over products	0.59 → 1.75	0.65 → 2.33
growth of $\lvert\varphi_j\rvert^2$ by purchase count	×2.2–3.7 in every band	×1.6–2.9 in every band
λ mean change	+0.65	+1.02
$\rho_0(k)$ change	up to +4.4	up to +15.3
product of whitening trace$/r$	×3.0 (40 whitenings)	×1.54 (29 whitenings)

Further measurements:

1. *Block swaps, synthetic.* Putting the household block back to step 242 gains +0.54 on held-out trips and loses 0.61 on training trips. The product-context block gains +0.30 and loses 0.69. Both together, back to step 420, reach −26.43.

2. *One whitening step* applied to the real step-360 fit changes the score by -0.007 ± 0.005 (held-out) and -0.010 ± 0.005 (training). The same step with $1.5M$ costs −1.3 and doubles the generated size.

3. *The estimate of M* is unbiased at $K = 256$: trace$/r$ is 1.020 at $K = 256$ and 1.019 at $K = 2{,}048$ (synthetic step 420). At $K = 16$ it is 1.037.

4. *The pooled posterior, real:* its mean has norm 0.21–0.23, its covariance eigenvalues lie in 0.80–1.16, and its trace$/r$ is 1.00.

5. *The bound gap* $\log p - L_8$ is the same for every basket size, at 0.14 nats at step 1,210.

3 Mechanism 1: memorisation by per-household and per-product parameters

Theory (`docs/overfitting_theory.pdf`). A block with d free numbers per household, fitted to that household's n_h trips, makes the training score better than the held-out score by about $d/(2n_h)$ per trip. With $d = 40$ and $n_h \approx 8$ that is about 2.5 nats. Early stopping stops this partway.

Why the hierarchical prior failed. `v4/hier_prior.py` centres each block column at its mean over rows. The base model had already fitted every row with early stopping on validation. A prior at the column mean pulls those validated row values toward the average, so it removes real signal (measured: −0.3 from the first step on). For γ and β, a single household's trips barely determine the values, so the EM update drives their prior scale to zero.

The fix that follows. The row values to keep are the base model's, $x^{(0)}$. Use the penalty

$$\frac{\tau}{2}\sum_{r,k}\bigl(x_{rk}-x^{(0)}_{rk}\bigr)^2 \quad \text{(in total log-likelihood units).} \tag{4}$$

For a row whose own likelihood has curvature a along a coordinate, the penalised fit moves that coordinate by the fraction $a/(a+\tau)$ of the unpenalised move. This is a smooth version of early stopping: $\tau = 0$ is the present fit and $\tau = \infty$ freezes the block. τ is chosen on validation. It is not a model of the population: $x^{(0)}$ was itself fitted to the training trips. It only limits how far the mixture fit moves these blocks.

Predictions. Synthetic: freezing or anchoring the household and product-context blocks removes most of the held-out loss after step 420; the best held-out score is at least the swap result -26.43. Real: this mechanism cannot explain the real loss, for two reasons. The training score fell as well, while memorisation raises it. And the real fits never trained these blocks.

4 Mechanism 2: the steps that are not gradient steps

The real fit loses on held-out *and* training trips. Steps 1–2 alone (stochastic gradient ascent on $\overline{L_K}$) can lose training likelihood only through noise or through the difference between L_K and $\log p$ (Section 5). Steps 3–4 are not gradient steps.

Whitening (parameter-expanded EM). Write $z = Az'$ with $z' \sim \mathcal{N}(0, I)$. Then $z \cdot \varphi_j = z' \cdot A\varphi_j$, and λ absorbs $-\frac{1}{2}|\varphi_j|^2$ exactly. So the new model with prior $\mathcal{N}(0, I)$ equals the old model with prior $\mathcal{N}(0, M)$, except for one term. In (2), the term $+\frac{1}{2}(n_c-1)_+|\psi_c|^2$ becomes $+\frac{1}{2}(n_c-1)_+|A\psi_c|^2$. It is linear in the number of extra products, and f_c (free only for $n_c = 2, 3, \geq 4$) and ρ_c (quadratic) cannot absorb it. A second omission: the EM update of a Gaussian prior also moves its mean, and $A = M^{1/2}$ folds the pooled mean $\bar{\mu}$ into the scale. *Measured:* at the real step-360 fit, one step costs 0.007–0.010, and $|\bar{\mu}| = 0.21$ shifts each product's utility by an amount with standard deviation 0.11. So whitening alone, at 29 steps, can account for at most about 0.3 of the 0.66, and only if nothing is recovered in between.

The reset of Adam's state. After a reset, Adam's first step moves every coordinate of φ, ψ by $\pm$ the learning rate, whatever the size of its gradient. With resets every 30 steps, this adds a random walk of about $0.003\sqrt{29} \approx 0.016$ per coordinate. That is far below the measured change ($|\Delta\varphi_j|^2 \approx 1.3$ after the best rotation). *Decided: not the cause.*

Moment matching. For λ_j, the gradient of the mean log-likelihood is $r_j - \hat{r}_j$: the data rate minus the model rate, averaged over the posterior. Dividing by r_j gives a damped Newton step, and its fixed point is the same as the gradient's. There are two departures:

1. it uses the gradient averaged over the last 30 steps, while Adam keeps moving the same parameters, so the two can overshoot together along directions where λ, ρ_0 and the scale of φ trade off;

2. its noise. A product bought c times in 7,680 trips gets a relative error of $1/\sqrt{c}$, so λ_j jitters with standard deviation about $0.35/\sqrt{c}$ around its target.

The size terms have the same structure. The measured change is a coordinated drift: $|\varphi|$ grows in every frequency band, λ rises by about 1, and ρ_0 rises by up to 15. That is the trade-off direction of the first departure.

Predictions. If moment matching drives the loss, a continuation without it loses less on training and held-out trips. If whitening drives it, a continuation without whitening does. Both continuations still train every parameter by the gradient, so they are fair tests.

5 Mechanism 3: training on the bound instead of the likelihood

The training gradient is

$$\nabla L_K = \sum_k \tilde{w}_k \, \nabla \log\big[\mathcal{N}(z_k; 0, I)\, p(S \mid z_k)\big], \qquad \tilde{w}_k = w_k \Big/ \sum_l w_l,$$

because the proposal q does not depend on the parameters. This is the self-normalised estimate of $\nabla \log p(S) = \mathbb{E}[\nabla \log p(S, z) \mid S]$. Its bias is of order $1/K$, and it points toward averages under q. Since q is wider than the posterior, the gradient behaves as if each posterior were wider than it is. *Measured:* at $K = 16$ the pooled second moment is overstated by 1.8% (1.037 against 1.019); at $K = 8$ it is about twice that. The fixed point of L_8 is therefore not the maximum-likelihood point, and the gap grows as posteriors narrow (synthetic: 0.06 → 0.14 nats).

Prediction. A continuation with $K = 32$ (bias four times smaller) loses less. This costs about 4 times as much per step, so it is the last experiment, run only if Sections 3–4 leave a loss unexplained.

6 Mechanism 4: generated size

At a stationary point of the likelihood, the size statistics match the data *averaged over each trip's posterior*:

$$\frac{1}{T} \sum_t \mathbb{E}\big[\, \mathbb{E}[n \mid z] \;\big|\; S_t \big] = \bar{n}_{\text{data}}.$$

Generation draws $z \sim \mathcal{N}(0, I)$. The two agree when the pooled posterior $\frac{1}{T}\sum_t p(z \mid S_t)$ equals $\mathcal{N}(0, I)$. That holds at the maximum-likelihood point of a *correctly specified* model, with large T. Whitening forces only the second moment to I. *Measured* (real): the mean is not zero ($|\bar{\mu}| = 0.22$), and the covariance has eigenvalues from 0.80 to 1.16.

Measured after the runs (`research/size_by_taste.py`, 512 synthetic training trips, 4 baskets per trip and way). Three facts settle this section.

1. The likelihood and the sampler agree. The mean number of categories implied by the gradient of $\log p$ with respect to ρ_0 (through the exact log-partition) is 7.874. The sampler at the same posterior draws gives 7.83–7.88. The data has 7.881. So the size statistic is matched under the posterior, as the stationarity condition requires.

2. Drawing z from $\mathcal{N}(0, I)$ instead gives 8.08–8.12 products (data 8.67).

3. Drawing z from a Gaussian with the *pooled posterior's own mean and covariance* gives 7.94–8.05, no better than $\mathcal{N}(0, I)$.

So the gap is not a matter of the first two moments, which whitening could fix. It comes from the *shape* of the pooled posterior. That posterior is a mixture of the trips' posteriors, not a Gaussian. The expected size grows faster than linearly in z, so a Gaussian with the right covariance still gives too few large baskets. This is misspecification of the taste distribution $\mathcal{N}(0, I)$, and no change to the training removes it. Fixing it needs a model change, such as a richer distribution of z, or drawing z from the household's own past posteriors for counterfactuals of known households. That choice is left to the user.

7 Mechanism 5: whitening and the training objective have different fixed points

This section was worked out after the first results (Section 9). Those results showed that on real data the *training objective itself* falls: from step 360 to the final checkpoint, $\overline{L_8}$ on 256 training trips went from -40.09 to -40.71, and $\log p$ from -39.93 to -40.44. So Mechanism 3, which needs L_8 to rise while $\log p$ falls, is ruled out. A gradient method does not lower its own objective unless other steps move the parameters away from it. This section shows that whitening does exactly that.

The scale direction. For $c > 0$ let T_c map the parameters to

$$\varphi_j \mapsto c\varphi_j, \quad \psi_c \mapsto c\psi_c, \quad \lambda_j \mapsto \lambda_j + \tfrac{1}{2}(c^2 - 1)|\varphi_j|^2,$$

and leave everything else fixed. By the change of variable $z = cz'$, the model $T_c\theta$ with prior $\mathcal{N}(0, I)$ is the model θ with prior $\mathcal{N}(0, c^2 I)$, apart from the $|\psi_c|^2$ term (Section 4).

Status after the experiments: the derivative in Proposition 1 is exact, but the claim that the prior's pull π explains the conflict was **refuted**. With the prior divided by 10, the moment still returns to 1.03–1.035 between whitenings (R3), and the synthetic fit is unchanged (S8). The match 1.044 against 1.035 was a coincidence.

Proposition 1. *Let $M = \frac{1}{T}\sum_t \mathbb{E}[zz^\top \mid S_t]$ be the pooled posterior second moment over the training trips. Then*

$$\frac{d}{dc}\Big[\frac{1}{T}\sum_t \log p(S_t; T_c\theta)\Big]_{c=1} = \operatorname{tr} M - r.$$

For the prior of the fit, $\frac{1}{2T}\big(200\sum_j |\varphi_j - \psi_{c(j)}|^2 + 20\sum_c |\psi_c|^2\big)$, the derivative along T_c at $c = 1$ is

$$\pi = \frac{1}{T}\Big(200\sum_j |\varphi_j - \psi_{c(j)}|^2 + 20\sum_c |\psi_c|^2\Big) > 0.$$

At a stationary point of the penalised likelihood, therefore, $\operatorname{tr} M = r + \pi$, and not r.

Proof. $\partial_c \log \mathcal{N}(z; 0, c^2 I) = |z|^2/c^3 - r/c$. Differentiating under the integral, the derivative of $\log p(S)$ at $c = 1$ is $\mathbb{E}[|z|^2 \mid S] - r$. The prior is a sum of squares that scales as c^2. □

Consequence. Whitening sets $M = I$, which is the stationary point of the likelihood *without* the prior. The gradient steps go toward the stationary point *with* the prior, where $\operatorname{tr} M/r = 1 + \pi/r$. So the two procedures have different fixed points, and the fit cycles between them without settling. Moreover, the gradient steps do not undo a whitening along T_c: they move φ, λ, ρ_0 and ρ_c separately. So each cycle leaves a remainder, and the remainders add up to the drift measured in Section 2.

The measured numbers fit. Real data: $\pi = 0.70$ per trip at step 360, which predicts that without whitening the fit settles at $\operatorname{tr} M/r \approx 1 + 0.70/16 = 1.044$. The run without whitening (R2) settles at 1.035 and stays there from step 390 to step 480. With whitening (R0), each block of 30 steps takes M from 1 back to 1.02–1.03. Synthetic: π grows from 0.36 (step 420) to 0.63 (step 1,210) as φ grows, so the gap that whitening closes widens as the fit goes on.

Two fixes that follow. The conflict exists only because the prior pulls along T_c, with strength π. Either remove one of the two procedures, or remove the pull.

(a) Train without whitening, so that every step serves one objective. Then, at the end, apply the EM update of the prior covariance once, with all else fixed (one whitening step).

(b) Keep whitening, and make the prior on φ weak enough that π/r is small. Dividing both precisions by 10 gives $\pi = 0.07$ on real data, so the two fixed points differ by $\operatorname{tr} M/r \approx 1.004$ instead of 1.044. With $T = 158{,}983$ trips, a prior of precision 2 on a loading still corresponds to very little information against the data.

Predictions. (i) Without whitening, the training objective does not fall, and the held-out score keeps improving past step 360. (ii) The final whitening step does not lower $\log p$ and moves the generated size toward the data. (iii) With the weaker prior and whitening on, the held-out score improves as in (i), and the generated size stays where whitening holds it.

8 Experiments

All runs use the existing code with three new switches: `--no-moment-match`, `--no-whiten`, and `--anchor BLOCKS --anchor-prec` τ (penalty (4); $\tau = \infty$ is written `--freeze`). Scores are paired, on 1,024 held-out and 1,024 training trips with $K = 256$. Selection is on held-out trips only; the test weeks are never opened.

Real (from the step-360 checkpoint, 480 steps to step 840, same batches). The real fits were made before the household, context and price blocks were trained: they trained only $\psi, \varphi, \lambda, \rho_0, \rho_c, f$, and did not re-estimate λ_c. The runs below keep that scope (`--interaction-only`). So on real data, Mechanism 1 is excluded from the start.

run	change	prediction if its mechanism is the cause
R0	none (control)	held-out and training both below step 360
R1	`--no-moment-match`	loss much smaller than R0 on both splits
R2	`--no-whiten`	loss much smaller than R0 on both splits

Synthetic (fresh 10-epoch fits, the same seed as the original run, which is the control).

run	change	prediction
S1	freeze household and product context	best held-out ≥ -26.43; small late loss
S2	anchor, $\tau = 10$	between S1 and the control
S3	anchor, $\tau = 100$	close to S1
S4	`--no-moment-match`	tells whether Mechanism 2 also acts here
S5	`--no-whiten`	likewise

The value $\tau = 10$ is about the measured likelihood curvature of a household coordinate with 2–4 trips (6–16), so it halves the movement of such a household.

Reading the results. For each run, report the held-out and training score at the best and final steps, against the control. Also report the generated size and the change in ρ_0 and $|\varphi|$. A mechanism is confirmed only if switching it off changes the late loss by more than twice the paired standard error. Otherwise it is rejected.

9 Results

Validation scores are on the fit's first 64 held-out trips ($K = 256$). Paired scores are on 512 held-out or training trips, with the same trips and random numbers for every checkpoint.

Synthetic, full fits (10 epochs).

run	change	best (step)	final, step 1,210
control	none	−28.34 (420)	−28.75
S1	household and product context frozen	**−27.24** (1,210)	**−27.24**
S2	anchored, $\tau = 10$	−28.20 (960)	−28.38
S3	anchored, $\tau = 100$	−27.54 (960)	−27.58
S4	no moment matching	−28.34	−28.72
S5	no whitening	−28.70 (720)	−28.98
S6	S1 without whitening	−27.37 (1,210)	−27.37
S8	S1 with the φ priors / 10	−27.27 (1,200)	−27.27
S9	S1, Adam state kept across whitening	**−27.23** (1,200)	−27.23

Mechanism 1 is confirmed. Freezing removes the late loss: the fit improves to the last step, and its best is 1.1 nats per trip above the control's best. The stronger the anchor, the better, and freezing is best. Moment matching has no effect (S4 equals the control). Without whitening, the synthetic fit is worse early on (S5). On the frozen recipe, whitening helps a little (S6 against S1). The prior strength does not matter (S8). Keeping Adam's state helps a little (S9). On synthetic data, with 522 products, whitening causes no late loss.

Real, continued from step 360 (interaction and size only).

run	change	step 480	step 600
R0	none (replays the original run exactly)	−35.898	−35.885
R1	no moment matching	−35.846	−35.828
R2	no whitening	−35.811	**−35.725**

Step 360 scores −35.744. R0 reproduces the original run exactly (−35.8983 at step 480, −35.8356 at 720, −35.8993 at 840), so the comparison is clean. Run R3 (φ priors / 10, whitening on) was stopped after 60 steps: its pooled moment returned to 1.030 and 1.035 after each whitening, the same as R0's 1.030 and 1.034, so Proposition 1's explanation (the prior) was refuted and the run gave no further information. R4 is whitening with the ψ compensation, and R5 is R4 with Adam's state kept.

paired, 1,024 trips, $K = 256$	held-out	training
R0 at step 840 − step 360	-0.27 ± 0.04	-0.23 ± 0.04
R2 at step 840 − step 360	$\mathbf{+0.26 \pm 0.02}$	$\mathbf{+0.30 \pm 0.04}$
R2 − R0	$+0.53 \pm 0.04$	$+0.54 \pm 0.04$

On real data, whitening causes the late loss. Without it, training past step 360 gains 0.26 nats per held-out trip and 0.30 per training trip, and the 64-trip validation score improves at every check (−35.744, −35.811, −35.725, −35.634, −35.583). The generated size of R2 falls, however: 6.53 against 7.26 in the data. R2 is the first real run that improves past step 360. On 512 paired held-out trips, R2 at step 600 scores −38.144, against −38.291 for step 360. The bound measurement (256 training trips, step 360 to the original final checkpoint) is in Section 7: L_8 fell by 0.61.

Final whitening (prediction (ii) of Section 7), on R2 at step 600, two EM steps of the prior covariance. Held-out: -0.007 ± 0.008. Training: $+0.004 \pm 0.009$. Generated size: 6.76 to 6.90 (data 7.22). The likelihood is unchanged, as predicted. The size moves the predicted way, but only by a third of the gap, as Section 6 explains.

The ψ term, one step (real step 360, 512 held-out trips, the same M): uncompensated -0.0068 ± 0.0049; compensated (`whiten_(..., compensate_psi=True)`) $+0.0012 \pm 0.0035$; their difference $+0.0081 \pm 0.0033$. For a step of $1.5M$: -1.32 ± 0.06 uncompensated, -0.24 ± 0.03 compensated (generated size 21.5 and 14.5). So the compensation makes whitening the neutral

step that theory says it should be. But over a run it is not enough: R4 at step 480 scores -35.901, the same as R0 (-35.898). Four whitenings at 0.008 each cannot account for the 0.09 that R0 loses against R2 in those 120 steps. Whitening must hurt through the training that follows it. R5 tests the reset of Adam's state.

Household taste in z (synthetic, 150 households, 1,196 trips). The share of the spread of trip posterior means $\mathbb{E}[z \mid S_t]$ that lies between households is -0.02 (S1) and -0.03 (control), that is zero. So a household mean for z would add nothing there.